

SAT Practice — Targeted Review

8 Custom Bluebook-Style Questions Based on Your Mistake Patterns

SECTION: MATH

Question 1

Advanced Math · Exponential Functions | Module 2 Hard

A researcher models the number of bacteria in a sample using the function

$$N(t) = 800(1.05)^{(3/5)t}$$

where $N(t)$ is the number of bacteria t hours after the start of the experiment. According to the model, the number of bacteria is predicted to increase by $p\%$ every 100 minutes. What is the value of p ?

- A) 0.83
- B) 3
- C) 5
- D) 60

Question 2

Advanced Math · Quadratic Discriminant | Module 2 Hard

$$2x(kx - 48) = -18$$

In the given equation, k is a positive integer constant. If the equation has no real solution, what is the least possible value of k ?

[Student-Produced Response]

Question 3

Problem-Solving & Data Analysis · Unit Conversion | Module 2 Hard

A pump delivers water at a rate that covers 540 square centimeters of surface area per second. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 100 centimeters.)

- A) 0.054
- B) 3.24
- C) 5.40
- D) 324

Question 4

Advanced Math · Rational Expressions | Module 2 Hard

Which expression is equivalent to

$$(y + 9)/(x - 5) + y(x - 5)/(x^2y - 5xy)$$

- A) $(xy^2 + 10xy - 5y) / (x^2y - 5xy)$
- B) $(xy^2 + 10xy - 5y) / (x^3y - 10x^2y + 25xy)$
- C) $(xy + 9y + 9) / (x^2y - 5xy + x - 5)$
- D) $(2y + 9) / (x^2y - 5xy)$

SECTION: READING AND WRITING

Question 5

Craft and Structure · Words in Context | Module 2 Hard

While some economists have praised the new trade agreement for creating a more level playing field, less _____ analysts caution that the agreement's provisions disproportionately benefit large multinational corporations at the expense of small domestic producers, potentially exacerbating inequality rather than reducing it.

- A) sanguine
- B) meticulous
- C) impartial
- D) forthright

Question 6

Standard English Conventions · Verb Forms | Module 2 Hard

A recent geological survey of the Mariana Trench, _____ that microbial communities thrive at depths exceeding 10,000 meters, has prompted marine biologists to reconsider long-held assumptions about the limits of life on Earth.

- A) revealing
- B) reveals
- C) revealed
- D) has revealed

Question 7

Standard English Conventions · Punctuation | Module 2 Hard

Excavations at Tell Brak in northeastern Syria have uncovered evidence that large-scale urban settlement in Mesopotamia may have begun 1,000 years earlier than was previously _____ discovery that has led some archaeologists to rethink the conventional model of urbanization in the ancient Near East.

- A) estimated; a
- B) estimated. A
- C) estimated, a
- D) estimated a

Question 8

Information and Ideas · Inference / Logical Completion | Module 2 Hard

The accuracy of climate models depends in part on how they represent cloud formation, a notoriously complex process. The Schmidt–Appleman model, widely used in atmospheric science, assumes that humidity is uniformly distributed within each grid cell of the atmosphere — an assumption necessitated by the fact that direct measurements of humidity at various points within individual grid cells are extremely sparse and inconsistent. Climate scientist Priya Nagarajan and colleagues note that while the model's temperature predictions align closely with satellite observations, it is therefore important to recognize that _____

- A) some discrepancies between the model's cloud-formation simulations and actual cloud patterns should be expected.
- B) the model's temperature predictions are likely to become less accurate as the effects of climate change intensify.
- C) although the model sometimes agrees with observational data, actual humidity levels are likely higher than those the model uses.
- D) discrepancies across the model's simulations of cloud patterns could be attributed to regional variations in humidity.